Statistics 280 — Practice Midterm 1C — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam; answers within rounding of these are fine.

Ex. 1 (30 points)

a (6 points)

(1) **Placebo effect**: believing you are being treated reduces symptoms or how you report them. (2) **Regression effect**: volunteers were recruited when their migraines were "worse than ever." Migraine frequency goes up and down; people selected at a peak tend to drift back toward their usual level whatever you do. (The natural course of the condition may also play a part.)

b (6 points)

Drug group vs placebo group, everyone as randomized: $6.5 - 5.0 =$ **1.5 fewer migraine days per month**. Randomization makes the groups comparable, and the placebo group experiences the same placebo and regression effects, so the difference isolates the drug.

c (6 points)

Adherers choose to adhere and differ from non-adherers (more health-conscious, fewer side effects, milder migraines). Look at the placebo group: adherers improved to 5.9 vs 7.8 for non-adherers *without any drug*, just as in the Clofibrate trial. Comparing drug adherers to the whole placebo group mixes the drug effect with the "adherer effect," so 2.3 overstates the benefit. Adherers vs adherers ($5.9 - 4.2 = 1.7$) is better but still observational: the drug itself (e.g. side effects) can change who adheres, so the two adherer groups may not be comparable. The clean comparison is the randomized one in b).

d (6 points)

Migraine days are self-reported and subjective. If patients knew, those on the drug would expect improvement (placebo effect) and placebo patients might report worse results or drop out. If doctors knew, they might encourage drug patients more, record borderline days differently, or give extra care, all of which bias the comparison in favor of the drug.

e (6 points)

Drug group: 8.0 to 5.0, an "effect" of **3 days**, double the real estimate of 1.5. A before/after design with historical controls has no contemporaneous control group, so it counts the placebo effect and the regression effect (the placebo group improved 1.5 days with no drug) as drug effect.

Ex. 2 (30 points)

a (8 points)

f is the sum of absolute deviations from the list 1, 4, 6, 6, 10, 10, 10 (weights mean repeats). It is minimized at the median, the 4th of 7 numbers: $u = 6$.

g is the sum of squared deviations from the same list, minimized at the mean: $(1 + 4 + 2 \cdot 6 + 3 \cdot 10)/7 = 47/7 \approx$ **6.71**.

b (6 points)
$h(u) = 4(u - 1.5)^2 + 4(u - 3.5)^2 + (u - 5)^2$: squared deviations from a list with four 1.5s, four 3.5s and one 5. Minimized at the mean: $(4 \cdot 1.5 + 4 \cdot 3.5 + 5)/9 = 25/9 \approx$ **2.78**.

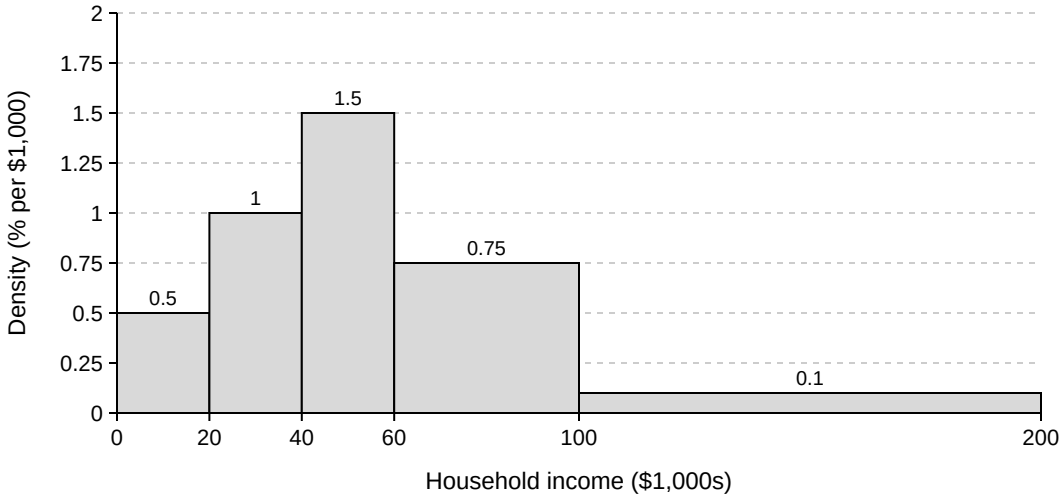
c (8 points)
Sum: $20 \times 50 = 1000$, corrected 1040, so the mean is **52**.
Sum of squares: $n(SD^2 + \text{mean}^2) = 20(100 + 2500) = 52{,}000$. Corrected: $52{,}000 - 50^2 + 90^2 = 57{,}600$. Mean of squares 2,880. Variance $= 2880 - 52^2 = 176$; SD $= \sqrt{176} \approx$ **13.27**.

d (3 points)
R gives SD^+. $SD = \sqrt{(n-1)/n}\, SD^+ = \sqrt{4/5} \times 4 \approx$ **3.58**.

e (5 points)
SD > IQR: 0, 0, 0, 0, 10. The 25th percentile is the 2nd number (need 1.25), 0; the 75th is the 4th (need 3.75), 0. IQR = 0. Mean 2, SD $= \sqrt{(4 \cdot 4 + 64)/5} = 4 > 0$.
IQR > SD: 0, 0, 10, 10, 10. 25th = 0, 75th = 10, IQR = 10. Mean 6, SD $= \sqrt{(2 \cdot 36 + 3 \cdot 16)/5} = \sqrt{24} \approx 4.90 < 10$. (Many other answers work.)

Ex. 3 (35 points)

a (8 points)
Blocks: [0,20) 10%, height $10/20 = 0.5$; [20,40) 20%, height 1.0; [40,60) 30%, height 1.5; [60,100) 30%, height 0.75; [100,200] 10%, height 0.1 (all in % per $1,000).



b (8 points)
Median: 30% below 40; we need 20% of the 30% in [40,60): $40 + 20 \cdot \frac{20}{30} =$ **53.33** thousand.
Mean $= 10(0.1) + 30(0.2) + 50(0.3) + 80(0.3) + 150(0.1) =$ **61** thousand. Mean > median because of the long right tail (a few incomes up to 200).

c (5 points)
[50,60): $1.5 \times 10 = 15\%$; [60,100): 30%; [100,120): $0.1 \times 20 = 2\%$. Total **47%**.

d (6 points)
25th: 10% below 20; need 15% of the 20% in [20,40): $20 + 20 \cdot \frac{15}{20} = 35$. 75th: 60% below 60; need 15% of the 30% in [60,100): $60 + 40 \cdot \frac{15}{30} = 80$. IQR = **45** thousand.

e (5 points)
The change is increasing, so percentiles carry over. Median $0.8(53.33) + 3 =$ **45.67**; mean $0.8(61) + 3 =$ **51.8**; IQR $0.8 \times 45 =$ **36** (adding 3 doesn't change spread).

f (3 points)
Mayor: the **median** (not pulled up by the few very high incomes). Tax office: the **mean**, since total = number of households × mean.

Ex. 4 (35 points)

a (4 points)
$z = 2$. Area within 2 is 95.45%, so $(100 - 95.45)/2 =$ **2.28%**.

b (6 points)
$z = -1$ and $20/15 = 1.33$. $68.27/2 + 81.64/2 = 34.14 + 40.82 \approx$ **74.96%** (about 75%). Interpolate in the table for 1.33: between 80.64 (1.30) and 82.30 (1.35).

c (6 points)
Top 15% means 70% in the middle: $z = 1.04$. Cutoff $= 100 + 1.04(15) =$ **115.6**.

d (6 points)
2.28% of all test takers score above 130, and all of them are admitted. Among the admitted 15%: $2.28/15 =$ **15.2%**.

e (5 points)
If the 75th percentile is q, at least 25% of the values are $\geq q$, and the rest are ≥ 0. So mean $\geq 0.25q$, giving $q \leq 20/0.25 =$ **80**. For the median, at least 50% are $\geq q$, so $q \leq 20/0.5 =$ **40** (the median is at most twice the mean). This is Markov's $m/(1 - q/100)$.

f (4 points)
Chebyshev: at most $1/1.5^2 = 44.4\%$ lie 1.5 or more SDs away, so at least **55.6%** lie within. For the normal curve it is 86.64%. Chebyshev is a guarantee for every list, so it is much weaker.

g (4 points)
No. 62 is $(62 - 50)/5 = 2.4$ SDs above the mean. Chebyshev: at most $1/2.4^2 = 17.36\% < 20\%$ of values can be that far from the mean.

Ex. 5 (30 points)

a (14 points)
(i) $\overline{xy} = r\,SD_x SD_y + \bar{x}\bar{y}$. Group 1: $0.4(2)(5) + 10(50) = 504$. Group 2: $4 + 15(60) = 904$.
(ii) $\bar{x} = 0.6(10) + 0.4(15) = 12$, $\bar{y} = 0.6(50) + 0.4(60) = 54$. $\overline{x^2} = 0.6(4 + 100) + 0.4(4 + 225) = 154$, so $SD_x = \sqrt{154 - 144} = \sqrt{10} \approx 3.16$. $\overline{y^2} = 0.6(25 + 2500) + 0.4(25 + 3600) = 2965$, so $SD_y = \sqrt{2965 - 2916} = 7$.
(iii) $\overline{xy} = 0.6(504) + 0.4(904) = 664$. cov $= 664 - 12(54) = 16$. $r = 16/(3.16 \times 7) \approx$ **0.72**. It rose because Group 2 is higher on *both* x and y: combining stretches the cloud along the same upward trend. (Compare the W25 height/weight question.)

b (6 points)
Plot 1: 0.3. Plot 2: 0.8. Its y-axis runs from −8 to 8, so the cloud looks flat; rescale so the SDs match and it is tight. Plot 3: −0.5. Plot 4: 0.95. (Actual: 0.3, 0.8, -0.5, 0.95.)

c (10 points)
(i) **False**: r measures only *linear* association (a U-shape can have $r = 0$). (ii) **False**: r becomes $-r$; r is always between −1 and 1. (iii) **True**: ecological correlations from averages remove individual scatter and usually overstate the association. (iv) **True**: restricting the range of x (attenuation) lowers r. (v) **False**: r has no such interpretation; points can all be off the line.

Ex. 6 (40 points)

a (6 points)
Slope $= 0.75 \times 300/8 = 28.125$ dollars per °F. Through $(75, 1200)$: $\hat{y} = 28.125x - 909.38$.

b (5 points)
$z = (87 - 75)/8 = 1.5$; predicted $z_y = 0.75(1.5) = 1.125$; $1200 + 1.125(300) =$ **$1,537.50**.

c (8 points)
RMS error $= \sqrt{1 - 0.75^2} \times 300 = \sqrt{0.4375} \times 300 \approx$ **$198.43**. Middle 90%: $z = \pm 1.65$. $1537.5 \pm 1.65(198.43) = 1537.5 \pm 327.4$: about **$1,210 to $1,865**.

d (7 points)
Residual $=$ actual $-$ predicted $= 1300 - 1537.5 =$ **−$237.50**. Within the strip, $z = -237.5/198.43 \approx -1.2$. Area within 1.2 is 76.99%, so the left tail is 11.5%: about the **11th–12th percentile** of 87°F days.
Among all days: $z = (1300 - 1200)/300 = 0.33$. Area within 0.33 is about 26%, so $50 + 13 =$ about the **63rd percentile**. Same sales figure, very different standing: it's a good day overall but a bad hot day.

e (4 points)
No. 20°F is about 6.9 SDs below the mean, far outside the data (extrapolation). The line predicts $28.125(20) - 909 = -346$ dollars, which is impossible; the linear trend doesn't hold there.

f (6 points)

r is unchanged at **0.75** (a linear change with positive multiplier). SD of x becomes $8 \times 5/9 = 4.44$°C, so the slope becomes $0.75 \times 300/4.44 =$ **$50.63 per °C** (= 28.125 × 9/5). The RMS error is unchanged at **$198.43**: it depends only on r and SD(y).

g (4 points)

(i) **True**: RMS $= \sqrt{1-r^2}\, SD_y$ for every data set, and $\sqrt{1-r^2} \leq 1$. (ii) **False**: with heteroscedastic (e.g. fan-shaped) data, one strip can be more spread out than the data as a whole. It is approximately true for football-shaped data.